Tingfeng Lan

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Research Interests

Keywords: LLM training and inference, model storage, I/O-efficient systems

I build systems across the LLM stack, including training, inference, and storage, with a focus on I/O bottlenecks. My research explores tensor-centric model storage, efficient offloading for LLM training, and workload-aware serving.

I am a Ph.D. candidate in the DS^2 Lab at the University of Virginia, advised by Prof. Yue Cheng, and work closely with Prof. Juncheng Yang at Harvard University.

Education

University of Virginia Sep 2024 – Present
Ph.D. in Computer Science, Advisor: Prof. Yue Cheng *VA, USA*

Sichuan University Sep 2020 – Jun 2024
B.Eng. in Computer Engineering, Advisor: Prof. Mingjie Tang *Sichuan, China*

Industry Experience

Cloudflare May 2026 – Aug 2026
Research Intern, Manager: Ivan Nikulin
*- Optimized GPU kernel memory access patterns for **Unweight**, a lossless LLM tensor compression system. Tailored optimizations to different tensor shapes and dimensions, improving kernel performance by over 40%.*

AntGroup AI Infra Sep 2023 – Jul 2024
Research Intern, Manager: Jian Sha
*- Designed and implemented **DLRover-RM** (VLDB'24), a resource-aware optimization system for large-scale recommendation-model training that improves resource utilization and reduces training cost in cloud environments.*
*- Designed and implemented **m-LoRA** (VLDB'25), a multi-tenant LoRA training framework that enables parallel multi-adapter fine-tuning via pipeline parallelism, reducing memory redundancy and improving training throughput.*

Publications

NSDI'27 **LatentStore: An Efficient Latent-First Storage System for AI-Generated Images**
Zirui Wang, Yunjia Zheng, Tingfeng Lan, Zhaoyuan Su, Haoran Ni, Juncheng Yang, Yue Cheng.
Accepted to the USENIX Symposium on Networked Systems Design and Implementation, 2027.

SIGMOD'27 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates**
Tingfeng Lan, Yusen Wu, Bin Ma, Zhaoyuan Su, Rui Yang, Tekin Bicer, Masahiro Tanaka, Olatunji Ruwase, Dong Li, Yue Cheng.
Accepted to SIGMOD 2027. Adopted into DeepSpeed and featured on PyTorch blog.

SOSP'26 **TensorDex: Rethinking AI Model Hub with Tensor-Centric Compression**
Tingfeng Lan, Zirui Wang, Yunjia Zheng, Zhaoyuan Su, Juncheng Yang, Yue Cheng.
In Proceedings of the ACM SIGOPS Symposium on Operating Systems Principles.
Distinguished Artifact Award.

WISE'26 **SCORPIO: Serving the Right Requests at the Right Time for Heterogeneous SLOs in LLM Inference**
Yinghao Tang, Tingfeng Lan, Xiuqi Huang, Hui Lu, Wei Chen.
Accepted to WISE 2026.

ACL Demo'26 **Demonstrating ViviDoc: Generating Interactive Documents through Human-Agent Collaboration**
Yinghao Tang, Yupeng Xie, Yingchaojie Feng, Tingfeng Lan, Jiale Lao, Yue Cheng, Wei Chen.
In Proceedings of the Annual Meeting of the Association for Computational Linguistics: System Demonstrations.

ACL'26 **IGenBench: Benchmarking the Reliability of Text-to-Infographic Generation**
Yinghao Tang, Xueding Liu, Boyuan Zhang, Tingfeng Lan, Yupeng Xie, Jiale Lao, Yiyao Wang, Haoxuan Li, Tingting Gao, Bo Pan, Luoxuan Weng, Xiuqi Huang, Minfeng Zhu, Yingchaojie Feng, Yuyu Luo, Wei Chen.

MLSys'26 **MorphServe: Efficient and Workload-Aware LLM Serving via Runtime Layer Swapping and KV Cache Resizing**
Zhaoyuan Su, Tingfeng Lan, Zirui Wang, Juncheng Yang, Yue Cheng.

MLSys'26 **λScale: Enabling Fast Scaling for Serverless Large Language Model Inference**
Minchen Yu, Rui Yang, Chaobo Jia, Zhaoyuan Su, Sheng Yao, Tingfeng Lan, Yuchen Yang, Yue Cheng, Wei Wang, Ao Wang, Ruichuan Chen.

ICDE'26 **DLRover-LM: LLM Pre-Training Framework with Thousands of Accelerators in AntGroup**
Ziling Huang, Zhengmao Ye, Qingsong Cai, Zelong Huang, Bo Sang, Haitao Zhang, Jian Sha, Tingfeng Lan, Hui Lu, Yuanchun Zhou, Mingjie Tang.
In Proceedings of the 42nd IEEE International Conference on Data Engineering.

NSDI'26 **ZipLLM: Efficient LLM Storage via Model-Aware Synergistic Data Deduplication and Compression**
Zirui Wang, Tingfeng Lan, Zhaoyuan Su, Juncheng Yang, Yue Cheng.
In Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation.

VLDB'25 **mLoRA: Fine-Tuning LoRA Adapters via Highly-Efficient Pipeline Parallelism in Multiple GPUs**
Zhengmao Ye*, Dengchun Li*, Zetao Hu, Tingfeng Lan, Jian Sha, Sicong Zhang, Lei Duan, Jie Zuo, Hui Lu, Yuanchun Zhou, Mingjie Tang.
In Proceedings of 51st International Conference on Very Large Data Bases

VLDB'24 **DLRover-RM: Resource Optimization for Deep Recommendation Models Training in the Cloud**
Qinglong Wang*, Tingfeng Lan*, Yinghao Tang, Bo Sang, Haitao Zhang, Jian Sha, Hui Lu, Ke Zhang, Mingjie Tang.
In Proceedings of 50th International Conference on Very Large Data Bases

Preprint **PathBee: Accelerating Shortest Path Querying via Graph Neural Networks**
Jiale Lao, Yinghao Tang, Tingfeng Lan, Mingjie Tang, Yuanchuan Zhou, Jianguo Wang.
In submission to ICDE'26.

* denotes equal contribution

Open Source Projects

DeepSpeed-ZenFlow: A stall-free offloading framework for LLM fine-tuning Oct 2024 - Present
Available on DeepSpeed, Received 40k+ ⭐ on GitHub
*- Designed and implemented **ZenFlow**, an importance-aware asynchronous offloading system that decouples GPU and CPU updates to eliminate GPU stalls. Achieved up to 5× end-to-end speedup, 2× reduction in PCIe traffic, and over 85% stall elimination while preserving accuracy.*

mLoRA: An efficient multi-tenant LoRA training system Sep 2023 - May 2024
Received 300+ ⭐ on GitHub
- Designed and implemented a training mechanism "BatchLoRA" which allows multiple LoRA adapters to share the pre-trained base model concurrently with reduced kernel launch overhead.

DLRover: An efficient autodl system with fault-tolerance awareness Jun 2023 - March 2024
Received 1.6k+ ⭐ on GitHub, Joined LF AI & Data Foundation ⚡
- Designed and implemented a hyper-parameter autotuner to optimize performance-relevant configurations, like micro-batch size, for maximum hardware utilization. Achieved over 95% memory utilization within a 30s estimation and re-configuration time; An elastic trainer, allowing for real-time hyper-parameter configuration during training sessions, thereby eliminating the restart overheads typically necessary in conventional training frameworks.

Funding and Grants

2025 **Modal Research Grant;** Total: $2,000 GPU credits; Led the proposal writing and submission;

Service & Activities

2027 **Reviewer for WWW'27**
2025-2026 **Shadow PC for EuroSys'26**
2025-2026 **Artifact Evaluation Committee for EuroSys'26, NSDI'26, OSDI'26**
2023–2024 **Journal Reviewer for IEEE TBD'24**

Talks

2025 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates;**
Invited talk: DeepSpeed Team
2025 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates;**
Poster presentation: UVA 2025 AI/ML Resource Fair

Awards

2026 **SOSP Distinguished Artifact Award** (TensorDex)
2026 **EuroSys Distinguished AEC Award**